```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\CoustomizePing> ^C
PS C:\CoustomizePing> & C:/Users/REC.DESKTOP-JAM6D74/AppData/Local/Programs/Python/P
==========
Customized Ping Utility
==========
==========
Enter Website or IP Address: google.com

Ping Results

Pinging google.com [142.251.221.142] with 32 bytes of data:
Reply from 142.251.221.142: bytes=32 time=25ms TTL=120
Reply from 142.251.221.142: bytes=32 time=29ms TTL=120
Reply from 142.251.221.142: bytes=32 time=9ms TTL=120
Reply from 142.251.221.142: bytes=32 time=5ms TTL=120

Ping statistics for 142.251.221.142:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 29ms, Average = 17ms

Server Status : Reachable
```

EXPERIMENT

DEVELOP A CUSTOMIZED PING COMMAND TO TEST THE SERVER CONSTANT L4

AIM:

To develop a customized Ping command in Python that checks the connectivity between a client and a server by sending ICMP Echo Request packets and displaying the response time and status.

REQUIREMENTS:

Hardware:

- ~~Computer~~ Laptop
- Internet connection

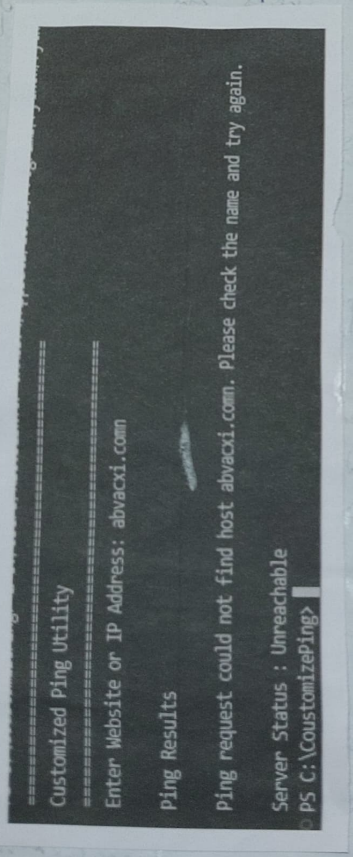
Software:

- Windows 10/11 or Ubuntu Linux
- Python 3.10 or above.
- Visual studio code (vs code).

Theory

PROCEDURE:

1. Install Python and VS code.
2. Create a python file named ping-PJ.
3. Import the required modules
4. Accept the hostname or IP address from the user.
5. Detect the operating system.
6. Use ping -n 4 or windows or ping -c 4 on Linux/mac OS.



7. Execute the ping command using the subprocess module.

8. Display the ping output, including reply time and packet statistics.

9. Check the return code to determine whether the server is reachable.

10. Display Reachable or Unreachable based on the result.

Result:

Customized ping command to test the server connectivity is verified successfully.